Examples

1. **The "Monty Hall" problem, revisited**

   You're on a game show, and are presented with three doors. One hides a car; the other two hide goats. You're allowed to choose a door. Monty Hall then opens one of the other two doors to reveal a goat. You are now allowed to either stick with the door you chose initially, or switch to the other closed door. What should you do (assuming that you are hoping to get the car, and not the goat): stick with your original choice, or switch?

   We consider two cases:

   (a) **Monty never opens the door with the car.**

   (b) **Monty chooses at random between the 2 doors that you didn't choose.**

   Let us call the doors A, B and C, and let us assume ("without loss of generality," as mathematicians like to say) that you choose door A. Moreover, let us initially *not* condition on Monty revealing the goat.

   There are six possible outcomes of the exeriment, { C$\boxed{\text{G}}$G, CG$\boxed{\text{G}}$, G$\boxed{\text{C}}$G, GC$\boxed{\text{G}}$, G$\boxed{\text{G}}$C, GG$\boxed{\text{C}}$ }. (The three letters denote the objects behind the three doors, so that CGG means the car is behind door A. The boxes indicate which door Monty opens. Since you chose door A, he won't be opening that door, but one of the other two.)

   In scenario (a), Monty always opens a door with a goat. Thus, if the door you chose hid the car (that is, if the car was behind door A), he'll choose at random between the doors B and C. On the other hand, if the car is behind door B, he'll always open door C, and vice versa. Thus, under scenario A, the events { C$\boxed{\text{G}}$G } and { CG$\boxed{\text{G}}$ } each have probability 1/6; events { GC$\boxed{\text{G}}$ } and { G$\boxed{\text{G}}$C } each have probability 1/3, and events { G$\boxed{\text{C}}$G } and { GG$\boxed{\text{C}}$ } each have probability 0.

   In scenario (b), in which Monty chooses at random between the two remaining doors, the six outcomes listed above each have probability 1/6.

   Now, define the events A = { the car is behind door A } = { C$\boxed{\text{G}}$G, CG$\boxed{\text{G}}$ }, B = { the car is behind door B } = { G$\boxed{\text{C}}$G, GC$\boxed{\text{G}}$ }, C = { the car is behind door C } = { G$\boxed{\text{G}}$C, GG$\boxed{\text{C}}$ }, and D = { Monty opens a door with a goat } = { C$\boxed{\text{G}}$G, CG$\boxed{\text{G}}$, GC$\boxed{\text{G}}$, G$\boxed{\text{G}}$C }.

   Our objective: we want to know the chance that the door we originally chose hides the car, *given that Monty has opened a door revealing a goat*. This is Pr(A | D). If this probability is $< 1/2$, we should switch doors; if it is $> 1/2$, we should stick with our first choice; if it is $= 1/2$, it doesn't matter whether we switch or stick.

   The event "A and D" (the intersection between A and D) is simply event A, and Pr(A) = 1/3 in either case (a) or (b), so Pr(A and D) = 1/3.

   In case (a), Pr(D) = 1, and so Pr(A | D) = 1/3. We should switch doors.

   In case(b), Pr(D) = 2/3, and so Pr(A | D) = 1/2. It doesn't matter what we do.

   The appropriate action to take depends on what Monty is doing!

2. **Suppose a test for HIV correctly gives a positive result, if a person is infected, with probability 99.5%, and correctly gives a negative result, if a person is not infected, with probability 98%.**

   (a) **Suppose that 0.1% of a population are infected with HIV. Consider drawing a person at random and testing him or her for HIV infection.**
   Calculate Pr(infected | test is positive).

   Let I = { the person is infected } and P = { the person tests positive }. From the information above, on the *sensitivity* and *specificity* of the test, we have Pr(P | I) = 0.995 and Pr(not P | not I) = 0.98. From this information, Pr(not P | I) = 1 – Pr(P | I) = 0.005 and Pr(P | not I) = 1 – Pr(not P | not I) = 0.02. Note further that Pr(I) = 0.001, and so Pr(not I) = 0.999.

   We seek to calculate Pr(I | P). We use Bayes's rule. (Why use Bayes's rule here? because we want to "turn around the conditioning." We want to write Pr(I | P) in terms of things like Pr(P | I).)

   $$\Pr(I \mid P) = \Pr(I)\Pr(P \mid I) \,/\, [\,\Pr(I)\Pr(P \mid I) + \Pr(\text{not } I)\Pr(P \mid \text{not } I)\,]$$
   $$= 0.001 \times 0.995 \,/\, (0.001 \times 0.995 + 0.999 \times 0.02) \approx 4.7\%$$

   (b) **Consider a person drawn from a high-risk group, so that they have, *a priori*, probability 30% of being infected.**
   Calculate Pr(infected | test is positive).

   In this case, we change Pr(I) = 0.3 and Pr(not I) = 1 – Pr(I) = 0.7.

   Thus, $\Pr(I \mid P) = 0.3 \times 0.995 \,/\, (0.3 \times 0.995 + 0.7 \times 0.02) \approx 96\%$

3. **Mendel, revisited.**

Mendel's peas had either purple or white flowers; flower color is due to a single gene, for which the purple allele (*A*) is dominant to the white allele (*a*).

We cross two pure-breeding lines (one purple and one white) to produce the F_1 hybrid. We self the F_1 and choose an F_2 seed at random. We grow and self the F_2 and choose two F_3 seeds at random.

Consider the following events.

$$
\begin{aligned}
\mathrm{P} &= \{\mathrm{F}_2 \text{ has purple flowers}\} \\
\mathrm{O} &= \{\mathrm{F}_2 \text{ is homozygous } AA\} \\
\mathrm{E} &= \{\mathrm{F}_2 \text{ is heterozygous } Aa\} \\
\mathrm{W} &= \{\mathrm{F}_2 \text{ has white flowers}\} \\
\mathrm{A}_1 &= \{\mathrm{F}_3 \text{ number 1 has purple flowers}\} \\
\mathrm{A}_2 &= \{\mathrm{F}_3 \text{ number 2 has purple flowers}\}
\end{aligned}
$$

(a) **Are A_1 and A_2 independent?**

A_1 and A_2 are *not* independent! A_1 and A_2 are *conditionally independent*, given the genotype of the F_2 plant. But, as we will see, in the absence of information about the F_2 parent's genotype, the flower color of the F_3 plants are not independent.

First, note that the events E, O and W are *mutually exclusive*, and that [Pr(E) + Pr(O) + Pr(W)] = 1.

$$
\begin{aligned}
\Pr(A_1) = \Pr(A_2) &= \Pr(A_1 \mid E)\Pr(E) + \Pr(A_1 \mid O)\Pr(O) + \Pr(A_1 \mid W)\Pr(W) \\
&= (3/4) \times (1/2) + (1) \times (1/4) + 0 \times (1/4) = 5/8 \approx 63\%.
\end{aligned}
$$

$$
\begin{aligned}
\Pr(A_1 \text{ and } A_2) = {} & \Pr(A_1 \text{ and } A_2 \mid E)\Pr(E) + \Pr(A_1 \text{ and } A_2 \mid O)\Pr(O) \\
& + \Pr(A_1 \text{ and } A_2 \mid W)\Pr(W) \\
= {} & (3/4)^2 \times (1/2) + 1 \times (1/4) + 0 \times (1/2) = 17/32 \approx 53\%.
\end{aligned}
$$

Thus $\Pr(A_2 \mid A_1) = \Pr(A_1 \text{ and } A_2) / \Pr(A_1) = (17/32) / (5/8) = 17/20 = 85\%$, which is considerably greater than $\Pr(A_2)$. And so, A_2 and A_1 are not independent.

(b) **Calculate $\Pr(A_1 \text{ and } A_2 \mid E)$.**

A_1 and A_2 are conditionally independent, given E, and so $\Pr(A_1 \text{ and } A_2 \mid E) = (3/4)^2 = 9/16 \approx 56\%$.

(c) **Calculate $\Pr(A_1 \text{ and } A_2 \mid P)$.**

$$
\begin{aligned}
\Pr(A_1 \text{ and } A_2 \mid P) &= \Pr(A_1 \text{ and } A_2 \mid E)\Pr(E \mid P) + \Pr(A_1 \text{ and } A_2 \mid O)\Pr(O \mid P) \\
&= (3/4)^2 \times (2/3) + 1 \times (1/3) = 34/48 \approx 71\%.
\end{aligned}
$$

(d) **Calculate $\Pr(A_1 \text{ and } A_2)$.**

$\Pr(A_1 \text{ and } A_2) = \Pr(A_1 \text{ and } A_2 \mid P) \approx 71\%$.